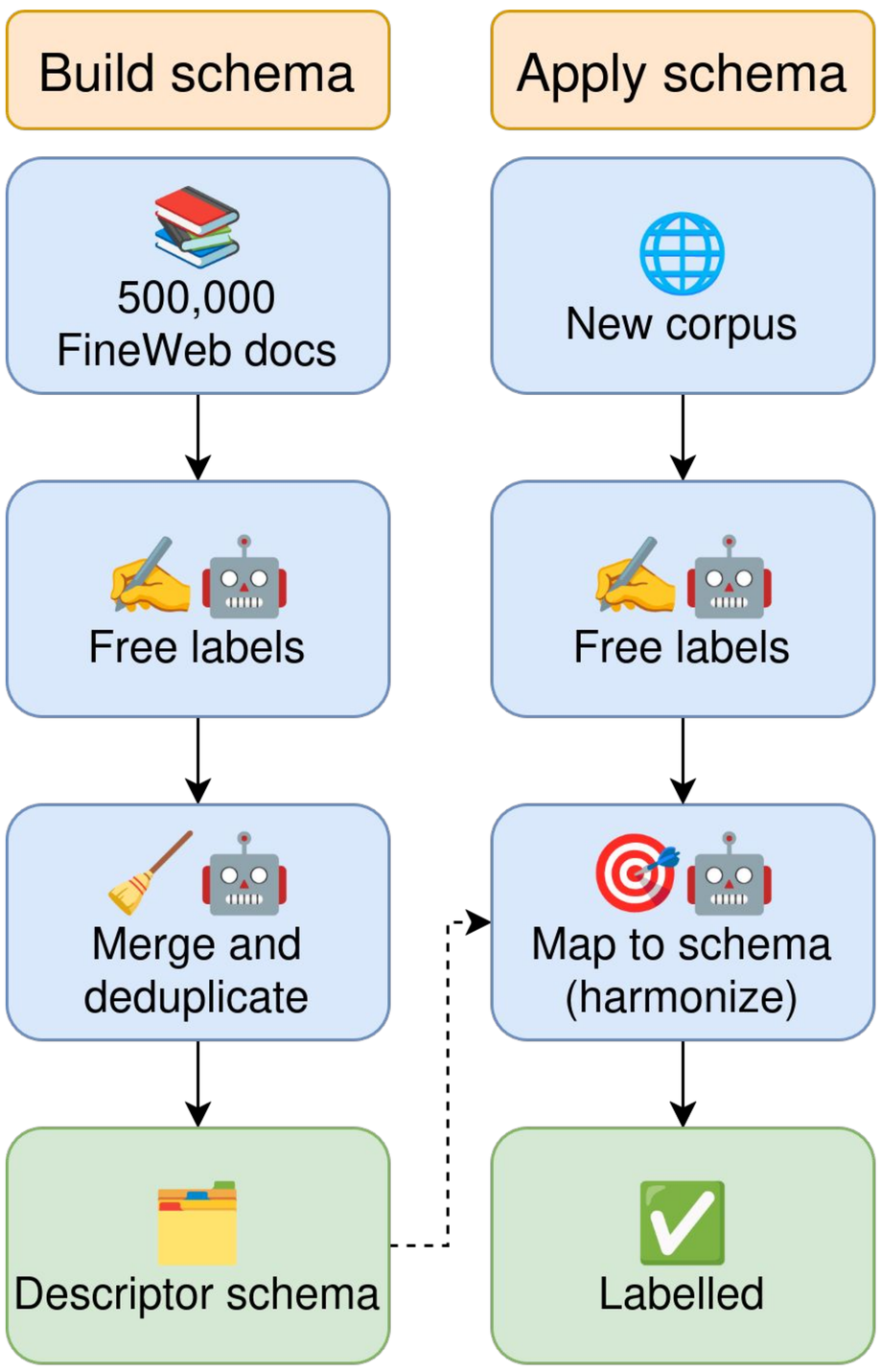
Task-Agnostic Web Document Annotation with LLM-Generated Descriptors

Otto Tarkka, Erik Henriksson, Jenna Kanerva, Filip Ginter
University of Turku, TurkuNLP
ohitar@utu.fi

Research Council of Finland

Web document descriptors

- LLM generated, task-agnostic document annotations
  -> 800,000 label annotation schema
- Describe the topic, sentiment, tone, style, and anything else that the model deems salient and important for that document
- Fine-grained and interpretable to support flexible downstream applications
- **Motivation**: Label once and use for anything; the task is not known upfront!



The problem

- Web corpora are huge but researchers often only need a small subset
  - **How do I find the needle in a haystack?**
  - e.g., sarcasm, formal language, step-by-step instructions
- Finding relevant documents is difficult and can require training custom classifiers
- Traditional annotation schemas are task-specific and often have coarse labels, thus only good for one task

descriptor	
tag	**definition**
formal tone and language	The formal tone and language used convey credibility, authority, and expertise, making it suitable for professional or official contexts.

Evaluation

- **Accuracy**: Raw descriptor 88.67%, harmonized 61.81% (LLM judge)
- **Aspects**: Topic is the most covered aspect (58% of descriptors) but style (20%), structure (10%) and quality (5%) are also present
- **Descriptors as interpretable features**: a simple linear classifier on descriptor tags matches TF-IDF on educational and format labels and its top features are readable phrases DEMO
- **Search**: A query for "sarcastic" documents found 5,459 documents, 74% judged relevant by an LLM (85% with raw descriptors)

Next steps and open questions

- Harmonization step costs accuracy; how could we avoid it?
  - LLMs are unable to reliably develop and maintain a large and growing labelset
- Better efficiency with a fine-tuned model
- Adjustable descriptor granularity and hierarchical structure
- Expand to multilingual corpora

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